Somaiya Vidyavihar University
K J Somaiya School of Engineering

Batch: **Roll No.:**
Experiment / assignment / tutorial No. 02
Grade: AA / AB / BB / BC / CC / CD /DD
Signature of the Staff In-charge with date

Title: Line Coding

Objectives:

1. **To understand different line coding techniques of base band signal**
2. **To develop a logic and generate waveforms of the given line coding techniques**

OUTCOME: analyze distortion less baseband transmission and reception of signal along with waveform coding techniques

Problem statement:

A. For the given binary sequence b = {1, 0, 1, 0, 1, 1}; sketch the waveforms representing the sequence b using the following line codes:

- **Unipolar NRZ**
- **Polar NRZ**
- **Bipolar NRZ**
- **Manchester**

Assume unit pulse amplitude and use binary data rate R_b = 1kbps. Write a program using MATLAB or any other open source software for the above line codes.

B. Determine and sketch the power spectral density (PSD) functions corresponding to the above line codes.
Hints: Use Rb = 1 kbps: Let f1 > 0 be the location of the first spectral null in the PSD function. If the transmission bandwidth B_T of a line code is determined by f1, determine BT for the line codes in question 1 as a function of Rb:
Let f_{P1}: first spectral peak; fn1: first spectral null; fp2: second spectral peak; fn2: second spectral null; such that all fp/n > 0

Record your observations in Table:

R_b =	F_{P1}	F_{N1}	F_{P2}	F_{N2}	B_T
Unipolar NRZ					
Polar NRZ					
Bipolar NRZ					
Manchester					

Expected output:

- **Waveforms of all line coding techniques**
- **Graph of magnitude response of each waveform**
- **Bandwidth required for transmitting each line code of the baseband waveform.**

Discussion/Conclusion:

Signature of faculty in-charge